Seminar Class 2 — Problems and Solutions

LM Monetary Economics and the Macroeconomy
Academic year 2026–2027

Exercise 1: A two-consumer endowment economy

Consider a two-period economy populated by two consumers, A (Ann) and B (Bob). Consumer $i \in \{A, B\}$ receives positive, perishable endowments Y_1^i and Y_2^i and chooses positive consumption C_1^i and C_2^i to maximize

$$U^i = \log C_1^i + \beta \log C_2^i.$$

Both consumers have the same discount factor $0 < \beta < 1$ and start with no financial assets. They can borrow or lend to each other at the common interest rate r, with $1 + r > 0$. Let S_1^i denote consumer i's saving; negative saving means borrowing. All loans are repaid in period 2, so the budget constraints are

$$C_1^i + S_1^i = Y_1^i, \qquad C_2^i = Y_2^i + (1+r)S_1^i.$$

Equivalently, $C_1^i + C_2^i/(1+r) = Y_1^i + Y_2^i/(1+r)$ (this is the intertemporal or lifetime budget constraint). The interest rate is determined in general equilibrium by asset-market clearing:

$$S_1^A + S_1^B = 0.$$

(a) Derive Ann's period 1 consumption function and show that

$$C_1^A(r) = \frac{1}{1+\beta}\left(Y_1^A + \frac{Y_2^A}{1+r}\right).$$

[This equation shows a general property of logarithmic preferences: expenditure shares on the available goods are constants that correspond to the relative weights on the logarithmic summands in U^i. Thus spending on period 1 consumption, C_1^i, is a fraction $1/(1+\beta)$ of lifetime income (measured in date 1 consumption units), and spending on period 2 consumption, $C_2^i/(1+r)$, is a fraction $\beta/(1+\beta)$ of lifetime income. This property follows from the unitary intertemporal substitution elasticity.]

(b) Derive Ann's saving function and show that

$$S_1^A(r) = Y_1^A - C_1^A(r) = \frac{\beta}{1+\beta}Y_1^A - \frac{1}{(1+\beta)(1+r)}Y_2^A.$$

(c) State the definition of a competitive equilibrium for this economy.

Hint: Your definition should specify Ann's consumption allocation (C_1^A, C_2^A), Bob's consumption allocation (C_1^B, C_2^B), and an interest rate r. What conditions must each consumer's choices satisfy when taking r as given? What condition must hold for their combined asset demand, $S_1^A + S_1^B$?

(d) Compute the equilibrium interest rate.

(e) How does faster growth in Bob's endowment affect Ann's welfare? Consider an increase in Y_2^B, holding Y_1^B, Ann's endowments, and β fixed. Observe that the resulting rise in Y_2^B/Y_1^B raises the equilibrium interest rate.

Define consumer i's *autarky interest rate*, r^i_{aut}, as the hypothetical rate at which that consumer would choose neither borrowing nor lending and would consume their own endowment in each period. It satisfies

$$1 + r^i_{\text{aut}} = \frac{Y_2^i}{\beta Y_1^i}.$$

This is an individual benchmark, not a separate market interest rate. Show that the derivative of Ann's maximized utility with respect to r is

$$\frac{dU^A}{dr} = \frac{\beta}{1+r}\left[\frac{r - r^A_{\text{aut}}}{(1+r) + \beta(1 + r^A_{\text{aut}})}\right].$$

What is your conclusion?

Solution to Exercise 1

(a) Consumption. Taking r as given, Ann chooses consumption to maximize utility subject to her lifetime budget constraint. Her Lagrangian is

$$\mathcal{L} = \log C_1^A + \beta \log C_2^A + \lambda\left(Y_1^A + \frac{Y_2^A}{1+r} - C_1^A - \frac{C_2^A}{1+r}\right).$$

The first-order conditions are

$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial C_1^A} &= \frac{1}{C_1^A} - \lambda = 0, \\
\frac{\partial \mathcal{L}}{\partial C_2^A} &= \frac{\beta}{C_2^A} - \frac{\lambda}{1+r} = 0, \\
\frac{\partial \mathcal{L}}{\partial \lambda} &= Y_1^A + \frac{Y_2^A}{1+r} - C_1^A - \frac{C_2^A}{1+r} = 0.
\end{aligned}$$

Eliminating λ gives the Euler equation,

$$\frac{1}{C_1^A} = \frac{\beta(1+r)}{C_2^A} \quad \Longrightarrow \quad C_2^A = \beta(1+r)C_1^A.$$

Substituting for C_2^A in the lifetime budget constraint gives

$$\begin{aligned}
C_1^A + \frac{\beta(1+r)C_1^A}{1+r} &= Y_1^A + \frac{Y_2^A}{1+r}, \\
(1+\beta)C_1^A &= Y_1^A + \frac{Y_2^A}{1+r}.
\end{aligned}$$

Dividing by $1+\beta$ yields

$$C_1^A(r) = \frac{1}{1+\beta}\left(Y_1^A + \frac{Y_2^A}{1+r}\right).$$

Strict concavity of logarithmic utility ensures that these conditions identify the unique optimum. The same argument applies to Bob.

(b) Saving. Using $S_1^A = Y_1^A - C_1^A$ and the result from part (a),

$$\begin{aligned} S_1^A(r) &= Y_1^A - \frac{1}{1+\beta}\left(Y_1^A + \frac{Y_2^A}{1+r}\right) \\ &= \frac{\beta}{1+\beta} Y_1^A - \frac{Y_2^A}{(1+\beta)(1+r)}. \end{aligned}$$

(c) Competitive equilibrium. Given endowments $Y_t^i > 0$ and the common discount factor β, a competitive equilibrium is a list $((C_1^A, C_2^A), (C_1^B, C_2^B), r)$, with $C_t^i > 0$ and $r > -1$, such that:

1. For each $i \in \{A, B\}$, the allocation (C_1^i, C_2^i) solves the consumer's problem, taking r as given:

$$\begin{aligned} &\max_{x_1, x_2 > 0} \ \log x_1 + \beta \log x_2 \\ &\text{subject to} \quad x_1 + \frac{x_2}{1+r} = Y_1^i + \frac{Y_2^i}{1+r}. \end{aligned}$$

2. The asset market clears:

$$S_1^A + S_1^B = 0, \qquad S_1^i = Y_1^i - C_1^i.$$

Given the individual budget constraints, asset-market clearing is equivalent to goods-market clearing in both periods. For period 1,

$$\begin{aligned} 0 = S_1^A + S_1^B &= (Y_1^A - C_1^A) + (Y_1^B - C_1^B), \\ C_1^A + C_1^B &= Y_1^A + Y_1^B. \end{aligned}$$

For period 2, adding the two flow budget constraints gives

$$\begin{aligned} C_2^A + C_2^B &= Y_2^A + Y_2^B + (1+r)(S_1^A + S_1^B) \\ &= Y_2^A + Y_2^B. \end{aligned}$$

These are goods-market clearing in periods 1 and 2. Conversely, given the individual budget constraints, period 1 goods-market clearing implies $S_1^A + S_1^B = 0$; period 2 goods-market clearing implies $(1+r)(S_1^A + S_1^B) = 0$, with the same conclusion since $1 + r > 0$.

(d) Equilibrium interest rate. Bob's saving function is

$$S_1^B(r) = Y_1^B - C_1^B(r) = \frac{\beta}{1+\beta} Y_1^B - \frac{Y_2^B}{(1+\beta)(1+r)}.$$

Adding the two saving functions and imposing asset-market clearing gives

$$0 = \frac{\beta}{1+\beta}(Y_1^A + Y_1^B) - \frac{Y_2^A + Y_2^B}{(1+\beta)(1+r)}.$$

Multiply by $(1+\beta)(1+r)$ and rearrange:

$$\begin{aligned} \beta(1+r)(Y_1^A + Y_1^B) &= Y_2^A + Y_2^B, \\ 1 + r &= \frac{Y_2^A + Y_2^B}{\beta(Y_1^A + Y_1^B)}. \end{aligned}$$

Therefore the equilibrium net interest rate is

$$r = \frac{Y_2^A + Y_2^B}{\beta(Y_1^A + Y_1^B)} - 1.$$

Positive endowments and $\beta > 0$ imply $1 + r > 0$, as required.

Define aggregate income growth as $G = (Y_2^A + Y_2^B)/(Y_1^A + Y_1^B)$, so $1 + r = G/\beta$. Higher growth makes consumers want to consume some of their greater future income today: at the initial interest rate, aggregate desired saving falls below zero, so r must rise to discourage borrowing and encourage saving, restoring market clearing. For example, if artificial intelligence (AI) raises future aggregate income relative to current income, this mechanism predicts a higher equilibrium real interest rate. Conversely, a higher β means consumers are more patient and want to save more at the initial interest rate; r must fall to reduce desired saving and restore market clearing.

(e) Ann's welfare. Holding Y_1^B, Ann's endowments, and β fixed, the result from part (d) implies

$$\frac{\partial r}{\partial Y_2^B} = \frac{1}{\beta(Y_1^A + Y_1^B)} > 0.$$

To determine how this higher interest rate affects Ann, write her maximized utility as

$$U^A(r) = \log C_1^A(r) + \beta \log C_2^A(r).$$

Her period 2 budget constraint is $C_2^A = Y_2^A + (1+r)(Y_1^A - C_1^A)$. Differentiating with respect to r while holding her endowments fixed gives

$$\frac{dC_2^A}{dr} = S_1^A - (1+r)\frac{dC_1^A}{dr}.$$

Hence

$$\begin{aligned}
\frac{dU^A}{dr} &= \frac{1}{C_1^A}\frac{dC_1^A}{dr} + \frac{\beta}{C_2^A}\frac{dC_2^A}{dr} \\
&= \left[\frac{1}{C_1^A} - \frac{\beta(1+r)}{C_2^A}\right]\frac{dC_1^A}{dr} + \frac{\beta S_1^A}{C_2^A} \\
&= \frac{\beta S_1^A}{C_2^A}.
\end{aligned}$$

The term in square brackets is zero by the Euler equation: the induced changes in Ann's optimal consumption choices cancel in the utility derivative.

To obtain the expression in the question, use $Y_2^A = \beta Y_1^A(1 + r_{\text{aut}}^A)$ to rewrite saving and period 2 consumption as

$$\begin{aligned}
S_1^A(r) &= \frac{\beta Y_1^A}{1+\beta}\frac{r - r_{\text{aut}}^A}{1+r}, \\
C_2^A(r) &= \frac{\beta Y_1^A}{1+\beta}\left[(1+r) + \beta(1+r_{\text{aut}}^A)\right].
\end{aligned}$$

Substituting these expressions yields

$$\frac{dU^A}{dr} = \frac{\beta S_1^A}{C_2^A} = \frac{\beta}{1+r}\frac{r - r_{\text{aut}}^A}{(1+r) + \beta(1+r_{\text{aut}}^A)}.$$

Bob's future endowment affects Ann's utility only through the equilibrium interest rate, so

$$\frac{dU^A}{dY_2^B} = \frac{dU^A}{dr}\frac{\partial r}{\partial Y_2^B}.$$

Because $\partial r/\partial Y_2^B > 0$, the sign of this welfare effect is the sign of Ann's saving. If $S_1^A > 0$ (or $r > r_{\text{aut}}^A$), Ann is a lender and benefits from a marginal increase in Bob's future endowment. If $S_1^A < 0$ (or $r < r_{\text{aut}}^A$), she is a borrower and is harmed. If $S_1^A = 0$ (or $r = r_{\text{aut}}^A$), she neither borrows nor lends, and the first-order welfare effect is zero. A higher interest rate improves the return on lending but raises the cost of borrowing. These statements describe the marginal effect at the initial equilibrium.